RAKESH MAHENDRAN

Gen AI Software Engineer · Full Stack Engineer

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PROFILE

Gen AI & Full Stack Engineer who builds practical web applications with LLMs and RAG systems integrated end-to-end. Comfortable across the stack — frontend, backend, retrieval pipelines, and LLM orchestration — with 2.5 years of production experience shipping AI features at enterprise scale. At my best in close-knit teams where ideas turn quickly into products. Driven by curiosity and continuous growth.

SKILLS

Programming Languages: Python, TypeScript, JavaScript

AI / LLM & RAG: LangChain, OpenAI, Azure AI Foundry, AWS Bedrock, Qdrant, Pinecone, RAG, Hybrid Search, Reciprocal Rank Fusion (RRF), Metadata Filtering, Function Calling, Tool Use, Multi-Agent Orchestration, GitAgent (gitclaw), Prompt Engineering

Evaluation & Observability: RAGAS, Custom Retrieval Eval Suites, Retrieval Quality Metrics (recall, latency, link relevance), Problem-Query Analysis

Backend: FastAPI, Django, Flask, Node.js, Express.js

Frontend: Next.js, React, Redux Toolkit, Tailwind CSS, Framer Motion, React Native

Databases: PostgreSQL, MongoDB, Redis, Elasticsearch

Cloud & DevOps: AWS (Lambda, S3, Bedrock), Azure (VMs, App Services, IaaS/PaaS), Docker, Vercel, GitHub Actions, CI/CD, Selenium

Tools & Platforms: GitHub, Bitbucket, Postman, VS Code, Jupyter Notebook

EXPERIENCE

Software Developer, *The Yellow Network* **Aug 2023 – Present**

NIFO – AI Co-Innovation Platform · *nifo.theyellow.network*

- Co-led B2B AI platform from prototype to production with one other engineer, owning Next.js + Redux frontend and Django backend modules end-to-end while working directly with the company founder on product direction.
- Built RAG pipeline using LangChain ConversationalRetrievalChain, Qdrant vector DB, and text-embedding-3-large; iterated chunking strategy through real-user testing with GPT-4o-mini for generation.
- Powered AI-driven innovation discovery and intelligent startup–enterprise matching via semantic retrieval across a 10,000+ startup profile knowledge base for clients evaluating co-innovation partnerships.
- Evolved the intelligence layer from NLP-to-SQL queries to RAG-based semantic retrieval after identifying limitations in structured-query approaches; pivoted to LangChain + Qdrant as AI capabilities matured.
- Designed end-to-end enterprise workflows — problem statements, questionnaires, challenge evaluation, and application pipelines — connecting enterprise clients with startup partners.
- Automated multi-source data ingestion with Selenium across web sources; managed Azure (IaaS + PaaS) deployments and CI/CD pipelines for the production environment.
- Platform showcased at the Ynfinity CXO meetup in Mumbai, an exclusive event for enterprise innovation leaders.

Stack: Django, Next.js, TypeScript, LangChain, Qdrant, OpenAI, PostgreSQL, Selenium, Azure, Redux

TIA – TVSSCS Intelligent Assistant (Client: TVS SCS) · *tvsscs.com*

- **Solo-built public-facing chatbot** embedded on the client website, handling visitor support queries and capturing sales-qualified leads with HubSpot CRM integration for downstream sales handoff.
- Designed serverless architecture on **AWS Lambda** with **Pinecone**; tuned cold-start behavior and memory footprint for chat-grade UX.
- Built a **hybrid retrieval pipeline** combining semantic search with targeted service queries, fused via **Reciprocal Rank Fusion (RRF)** for ranked merging and improved recall on core service questions.
- Implemented **LLM-based metadata filter extraction** — a fast model translates natural language queries to structured filters (category, document_type, fiscal_year, etc.) applied as Pinecone metadata filters for precision retrieval.
- **Two-model architecture for cost/latency optimization** — fast model for filter extraction and intent detection, capable model for final answer synthesis.

- Built a **custom retrieval evaluation suite (150-query test set)** tracking success rate, response latency, link relevance, and category coverage; used problem-query analysis to iteratively improve retrieval accuracy.

Stack: Python, AWS Lambda, Pinecone, AWS Bedrock, HubSpot CRM

TVS Sidekick – Enterprise Multi-Agent MS Teams Chatbot (Client: TVS SCS)

- Contributed to a **multi-agent, function-calling production AI system** for the enterprise MS Teams chatbot — orchestrating multiple foundation models via **AWS Bedrock with tool use**, serving 100+ TVS SCS staff across support, warehouse, and IT.
- **Owned RFI Bot feature** end-to-end, integrating it into the agent platform's LLM-based intent classifier and the broader function-routing layer that orchestrates 8+ retrieval tools — SharePoint, Azure DevOps, intranet, code repository, Snowflake Cortex (NL-to-SQL), TVS website, and document store.
- Added **Redis caching** for frequent embedding lookups and integrated **AWS Bedrock** for scalable LLM inference; reduced repeated query latency on cached intents and improved response time for high-frequency tool calls.
- **Containerized the multi-service production stack** with Docker Compose — Flask API + Node.js Teams bot + Python NLP gateway + Python OCR gateway + Elasticsearch + PostgreSQL — simplifying local dev and production deployment.
- Shipped enhancements and bug fixes across the document-processing and intent-routing layers, improving response accuracy for common warehouse, support, and IT queries.

Stack: Python, Flask, Node.js, AWS Bedrock, Redis, Elasticsearch, PostgreSQL, Docker, Docker Compose, OpenTelemetry

INTERNSHIP

Full Stack Developer Intern, *Travelfika* **May 2022 – Aug 2023**

Travelfika – Travel Planner & Booking Platform · *travelfika.com*

- Built full-stack applications using React for the frontend and Node.js with MongoDB for the backend, delivering dynamic and responsive functionality across the booking platform.
- Built and integrated core booking modules (flight booking, seat selection, hotel, car, cruise) leveraging third-party travel APIs with a backend orchestration layer for reliable booking operations.
- Designed and implemented a secure authentication and onboarding system using React Hook Form with validation, JWT tokens, and session management for safe user sign-up and login flows.
- Developed a cross-platform mobile app using React Native with Expo, delivering a unified booking and itinerary experience across iOS and Android alongside the web app.
- Integrated GPT API for automated hyperlink content extraction and AI-generated day-by-day travel itineraries, enriching attraction descriptions and personalizing trip packages across flights, hotels, cars, and cruises modules.

Stack: React, React Native, Expo, Node.js, Express.js, MongoDB, GPT API, Redux, JWT

PROJECTS

PORTFOLIO COUNCIL – MULTI-AGENT AI INVESTMENT BOARD ON GIT (LYZR GITAGENT HACKATHON) · *GitHub* · *Demo*

- Built a **5-agent AI investment-research board** on Lyzr's **GitAgent** runtime — Onboarding, Analyst, Strategist, Risk Officer, and Execution agents run a structured rebalancing debate where the Risk Officer issues APPROVE / AMEND / VETO verdicts and loops weak proposals back to the Strategist.
- Made **git the agents' shared memory and governance layer** — every analysis, proposal, and verdict is a versioned, revertable commit, and a pre-commit hook blocks any rebalance the Risk Officer hasn't APPROVED.
- Built Python **tool-skills** the agents call during reasoning (analyze-holdings, check-market, import-holdings); shipped a **Next.js** dashboard + **FastAPI** bridge streaming the live run, provider-agnostic across AWS Bedrock / OpenAI / Anthropic / Azure.

Stack: Next.js, TypeScript, FastAPI, Python, GitAgent (gitclaw), AWS Bedrock, Docker

FLAKERSSTUDIO – AGENTIC CHATBOT BUILDER FOR WEBSITES & WORDPRESS · *github.com/RakeshMahendran/Flakers-studio*

- Built an agentic platform that ingests website or WordPress content and produces a fully-indexed, governance-checked chatbot ready for deployment — automating content fetching, chunking, embedding, and per-tenant policy configuration.
- Architected RAG pipeline with multi-tenant Qdrant isolation so each chatbot's vector data is fully separated; audit logging on every retrieval and generation decision.
- Focused on governance: built a server-side policy layer — intent filtering, source attribution, confidence thresholds — enforced before any AI response leaves the backend so the frontend cannot bypass the rules.
- Designed for enterprise readiness: every AI response auditable, every source attributable, every decision policy-checked before delivery.

Stack: Next.js, TypeScript, FastAPI, PostgreSQL, Qdrant, Azure OpenAI, LangChain

EDUCATION

B.E. Mechanical Engineering, *Government College of Technology, Coimbatore* **| CGPA: 8.3** **2020 – 2024**